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ECON 312

DATA SCIENCE FOR ECONOMIC ANALYSIS

Customer Churn Analysis

This report presents a complete statistical analysis of a telecommunications customer churn dataset. Starting from a raw file of 1599 customer records, a fixed random sample of 1000 customers was drawn using `set.seed(52)` followed by `slice_sample(n=1000)`. Three of those 1,000 records had a missing TotalCharges value and were removed, leaving a final analytic sample of 997 customers. Because no further `set.seed()` call occurs anywhere later in the script, every subsequent random step, including the train/test split in Section 4, continues deterministically from that same seed-52 stream: re-running the script from top to bottom will always reproduce the exact same sample, model fits and split reported here.

All statistical modeling, hypothesis testing and visualization were performed in R. The analysis is organized into four sections: (1) descriptive statistics and exploratory data analysis; (2) a multiple linear regression model explaining monthly charges; (3) a logistic regression model predicting customer churn; and (4) a validation exercise benchmarking logistic regression against Linear Discriminant Analysis (LDA) & linear regression against K-Nearest-Neighbors (KNN) regression, using an explicit 70/30 train-test split.

1. Descriptive Statistics & Data Exploration

1.1 Percentage Frequency Tables

Table 1 summarizes the distribution of four key categorical variables: churn status, contract type, internet service type, and payment method.

Variable	Category	Frequency (n)	Percentage (%)
Churn	No	739	74.12
	Yes	258	25.88
Contract	Month-to-month	566	56.77
	One year	209	20.96
	Two year	222	22.27
Internet Service	DSL	333	33.40
	Fiber optic	472	47.34
	No	192	19.26
Payment Method	Electronic check	325	32.60
	Bank transfer (auto.)	238	23.87
	Credit card (auto.)	224	22.47
	Mailed check	210	21.06

Table-1: Percentage frequency distribution of key categorical variables ($N = 997$)

Roughly one in four customers (25.88%) has churned. Month-to-month contracts dominate the customer base (56.77%), a segment generally associated with materially higher churn. Fiber-optic subscribers form the largest internet-service segment (47.34%), ahead of DSL (33.40%), while about one in five customers subscribes to no internet service at all. Electronic check is the most common payment method, used by roughly a third of customers.

1.2 Scatter Plot: Monthly Charges vs Tenure

Figure 1 plots Monthly Charges against Tenure, with points colored by churn status.

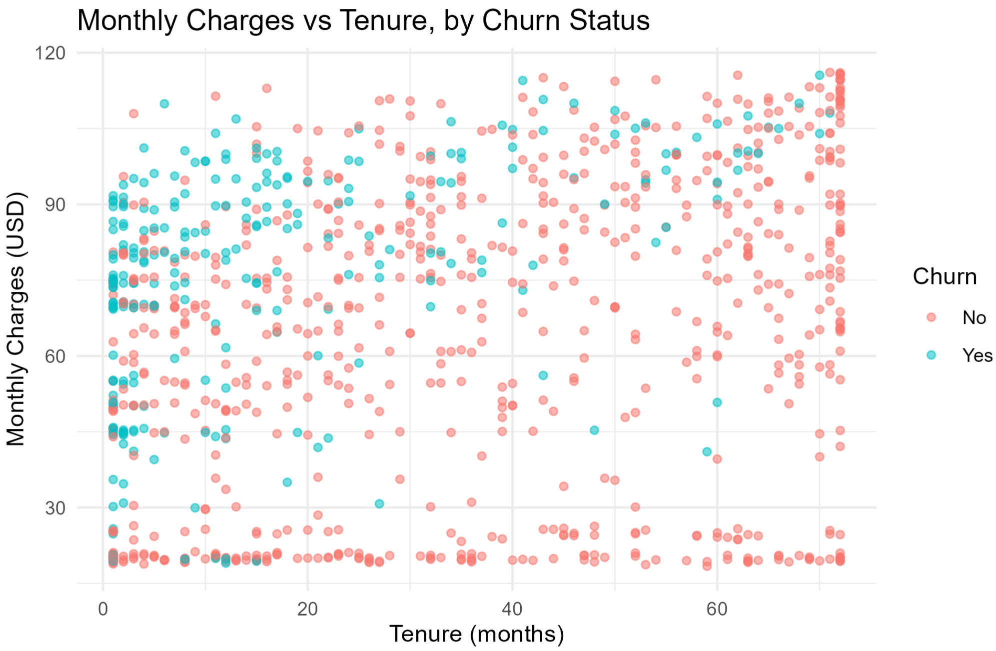

Figure-1: Monthly charges vs tenure, colored by churn status

The plot shows a weak positive linear association between tenure and monthly charges (Pearson $r = 0.261$), consistent with gradual add-on accumulation rather than a strong underlying trend. Churned customers are visibly concentrated in the lower-tenure region and are disproportionately represented among higher monthly-charge brackets, suggesting new customers facing steep early bills are at elevated risk of attrition.

1.3 Box Plot: Distribution & Outlier Detection

Figure 2 displays box plots of Monthly Charges stratified by contract type.

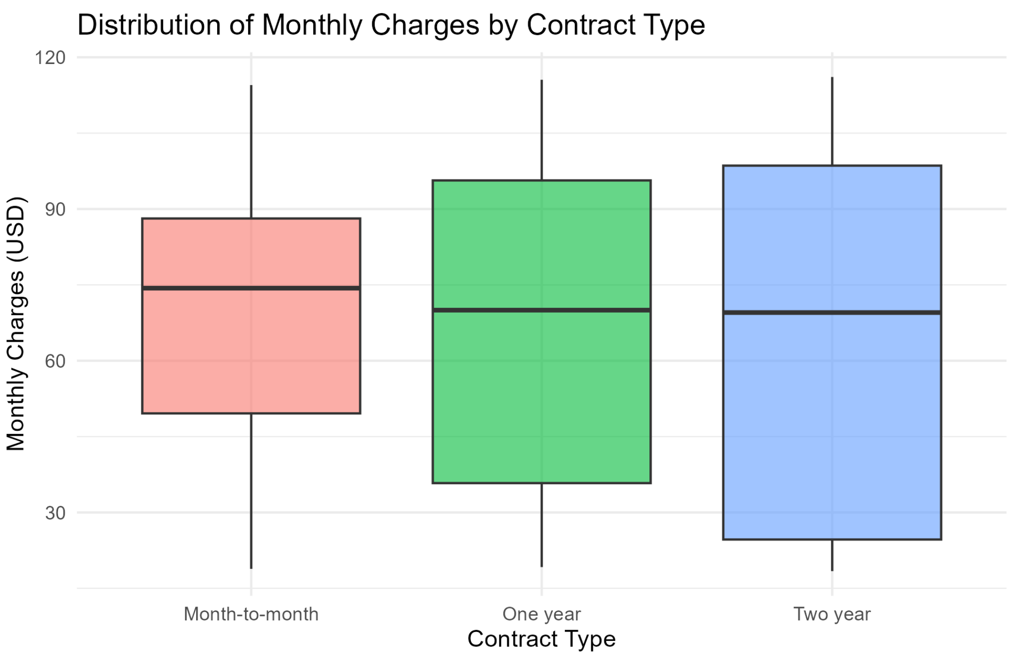

Figure-2: Distribution of monthly charges by contract type.

Interquartile ranges overlap substantially across contract tiers, indicating that contract duration alone does not sharply stratify bill size. An IQR-based check (1.5 IQR beyond $Q1/Q3$) found no outliers in tenure, monthly charges or total charges. So all subsequent modeling uses the complete 997-customer sample without trimming. Total Charges is, as expected, right-skewed (mean $2290.94 vs median $1416.75), since it is mechanically the running product of tenure and monthly charges ($r = 0.842$ with tenure; $r = 0.642$ with monthly charges).

2. Linear Regression Model & Statistical Inference

The first modeling task addresses a continuous outcome: Monthly Charges (USD). A multiple linear regression model was fit using tenure and four service/contract characteristics: contract type, streaming-TV subscription, paperless billing, and senior-citizen status.

2.1 Model Specification

$$\begin{aligned}
\text{MonthlyCharges} = \beta_0 &+ \beta_1\text{tenure} + \beta_2\text{Contract}_{\text{One year}} \\
&+ \beta_3\text{Contract}_{\text{Two year}} + \beta_4\text{StreamingTV}_{\text{No internet}} \\
&+ \beta_5\text{StreamingTV}_{\text{Yes}} + \beta_6\text{PaperlessBilling}_{\text{Yes}} \\
&+ \beta_7\text{SeniorCitizen}_{\text{Yes}} + \varepsilon
\end{aligned}$$

Reference categories: month-to-month contract, no streaming TV, non-paperless billing, and non-senior-citizen status.

2.2 Coefficient Estimates & Interpretation

Term	Estimate	Std. Error	t value	p value
Intercept	58.877	1.148	51.275	< 0.001
tenure	0.250	0.030	8.236	< 0.001
Contract: One year	-3.392	1.460	-2.324	0.020
Contract: Two year	-6.722	1.816	-3.701	< 0.001
Streaming TV: No internet service	-43.054	1.446	-29.779	< 0.001
StreamingTV: Yes	19.402	1.147	16.918	< 0.001
PaperlessBilling: Yes	3.234	1.063	3.044	0.002
SeniorCitizen: Yes	5.456	1.326	4.113	< 0.001

Table-2: OLS coefficient estimates for the Monthly Charges model ($df = 989$).

Every predictor is statistically significant. Holding all else constant, each additional month of tenure is associated with a $0.25 increase in monthly charges ($p < 0.001$). Relative to a month-to-month plan, one-year contracts are associated with $3.39 lower monthly charges ($p = 0.020$) and two-year contracts with $6.72 lower charges ($p < 0.001$). Streaming TV is the largest single driver of bill size: subscribing adds $19.402 per month, while having no internet service at all lowers the bill by $43.054 (both $p < 0.001$). Paperless billing and senior-citizen status are each associated with significant increases of $3.23 and $5.46 respectively ($p = 0.002$ and $p < 0.001$).

2.3 Overall Model Significance (F-test)

The overall F-test yields $F(7, 989) = 381.58$, $p < 0.001$, decisively rejecting the null hypothesis that all slope coefficients are simultaneously zero. The predictor set as a whole is highly statistically significant in explaining variation in monthly charges.

2.4 Goodness of Fit

The model achieves an R^2 of 0.7298 and an Adjusted R^2 of 0.7279, meaning approximately 73% of the variation in monthly charges is explained by tenure and the four service/contract attributes. The residual standard error is 15.35 on 989 degrees of freedom. Figure 3 plots residuals against fitted values.

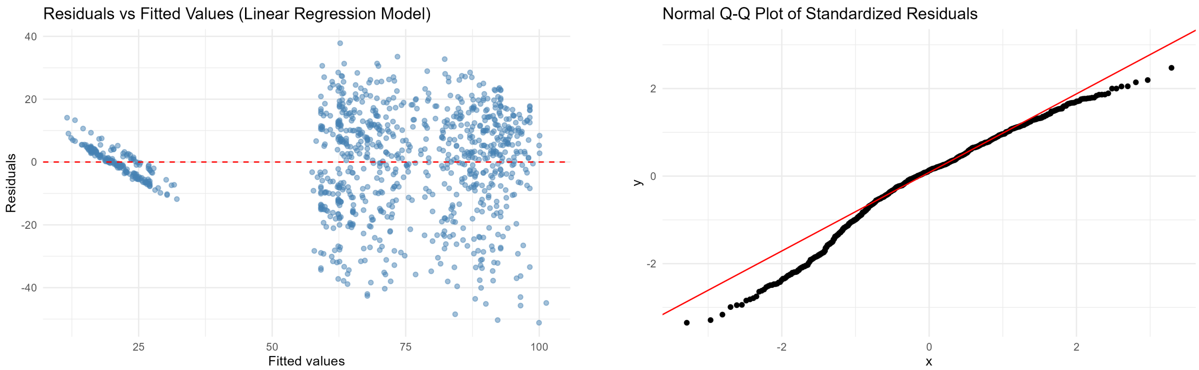

Figure-3: Residuals vs fitted values for the linear regression model & Normal Q-Q Plot of Standardized Residuals

A distinct diagonal band appears at low fitted values, corresponding to the "no internet service" segment whose bills cluster tightly near a fixed tier price; the remaining, larger cluster of points at higher fitted values shows a roughly symmetric scatter with no strong funnel shape. A variance inflation factor (VIF) check confirmed no problematic multicollinearity, with all values well below the conventional threshold of 5 (tenure: 2.28; Contract, one year: 1.49, two year: 2.42; StreamingTV, no internet: 1.38, yes: 1.33; PaperlessBilling: 1.14; SeniorCitizen: 1.06).

3. Logistic Regression Model & Statistical Inference

The second modeling task addresses a binary outcome: whether a customer churns (Yes/No). A logistic regression model was fit using tenure, monthly charges, contract type, internet service type, paperless billing, senior-citizen status, partner status and dependents status as predictors.

3.1 Coefficient Estimates, Log-Odds & Odds Ratios

Term	Estimate (log-odds)	Std. Error	z value	p value	Odds Ratio
Intercept	-0.557	0.439	-1.268	0.205	0.573
tenure	-0.035	0.006	-5.681	< 0.001	0.966
MonthlyCharges	-0.003	0.008	-0.367	0.714	0.997
Contract: One year	-0.563	0.276	-2.039	0.041	0.570
Contract: Two year	-1.807	0.567	-3.189	0.001	0.164
InternetService: Fiber optic	1.167	0.333	3.507	< 0.001	3.211
InternetService: No	-1.280	0.441	-2.905	0.004	0.278
PaperlessBilling: Yes	0.591	0.195	3.036	0.002	1.805
SeniorCitizen: Yes	0.262	0.217	1.212	0.225	1.300
Partner: Yes	0.032	0.210	0.152	0.880	1.032
Dependents: Yes	-0.319	0.242	-1.317	0.188	0.727

Table-3: Logistic regression coefficients, Wald tests and odds ratios for Churn = Yes ($df = 986$).

Each additional month of tenure multiplies the odds of churn by 0.966 (a 3.4% reduction per month, $p < 0.001$), a strong, highly significant protective effect. Contract duration is the most powerful retention lever: relative to month-to-month customers, one-year contract holders have 43% lower odds of churning ($OR = 0.570$, $p = 0.041$) and two-year contract holders have 83.6% lower odds ($OR = 0.164$, $p = 0.001$). Fiber-optic customers carry 3.21 times the odds of churning relative to DSL customers ($p < 0.001$) while customers with no internet service have 72.2% lower odds ($p = 0.004$). Paperless billing is associated with 80.5% higher odds of churn ($p = 0.002$). Monthly charges ($p = 0.714$), senior-citizen status ($p = 0.225$), partner status ($p = 0.880$) and dependents status ($p = 0.188$) are statistically insignificant predictors of churn once tenure and contract/service variables are controlled for.

3.2 Likelihood Ratio Test & Wald Tests

The null deviance is 1140.12 (996 df) and the residual deviance is 837.64 (986 df), yielding a likelihood-ratio chi-square statistic of 302.48 on 10 degrees of freedom, $p < 0.001$. The predictor set jointly improves substantially on the intercept-only model. Individually, the Wald z-tests in Table 3 show tenure, both contract terms, both internet-service terms and paperless billing are each significant at the 5% level, while monthly charges, senior-citizen status, partner, and dependents are not.

3.3 Goodness of Fit

The residual deviance of 837.64 represents a 26.5% reduction from the null deviance of 1140.12, corresponding to a McFadden pseudo R^2 of 0.2653, a good fit by conventional standards for binary-choice models. The model's AIC is 859.64.

4. Model Comparison

To assess genuine predictive accuracy rather than in-sample fit, the cleaned dataset ($N = 997$) was randomly partitioned into a training set of 697 customers (70%) and a held-out test set of 300 customers (30%). Because this split is drawn by the same `sample()` call continuing the seed-52 stream, it is fully reproducible: re-running the whole script will always yield this exact 697/300 split. All models were re-fit on the training data only and evaluated exclusively on the unseen test set.

4.1 Logistic Regression vs LDA

Both the logistic regression model and a Linear Discriminant Analysis (LDA) model (identical predictor set) were trained on the 697-customer training set and applied to the 300-customer test set at a 0.5 classification threshold.

Model	**Predicted**	**Actual: No**	**Actual: Yes**
Logistic Regression	No	203	41
	Yes	21	35
Linear Discriminant Analysis	No	202	37
	Yes	22	39

Table-4: Confusion matrices for Logistic Regression and LDA on the 300-observation test set.

Metric	Logistic Regression	LDA
Overall Accuracy	79.33%	80.33%
False Positive Rate	9.38%	9.82%
False Negative Rate	53.95%	48.68%

Table-5: Predictive accuracy comparison: Logistic Regression vs. LDA.

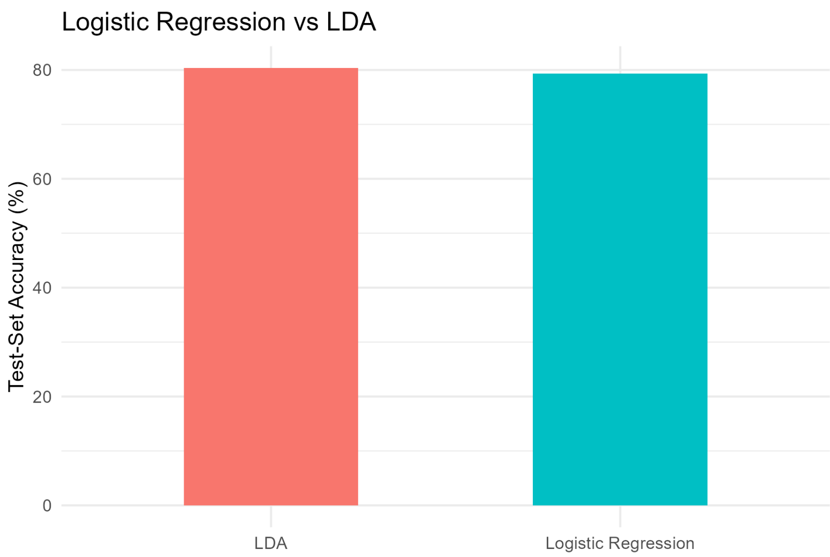

Figure-4: Test-set classification accuracy: Logistic Regression vs LDA.

On this particular train/test split, LDA outperforms logistic regression, the reverse of what was observed on the full raw dataset. LDA achieves 80.33% overall accuracy vs 79.33% for logistic regression and more importantly, a notably lower false negative rate (48.68% vs 53.95%) and LDA catches more of the actual churners, which matters most for a retention program, at the cost of a marginally higher false positive rate (9.82% vs 9.38%). This reversal illustrates that with a modestly sized test set (300 observations, only 76 of them churners), which of the two similar linear classifiers comes out ahead can depend on the specific train/test split; the difference here is well within the range attributable to sampling variability rather than a fundamental difference in modeling approach.

4.2 Linear Regression vs KNN Regression

The linear regression model from Section 2 was compared against a K-Nearest-Neighbors (KNN) regression algorithm for predicting Monthly Charges, using the identical (standardized, dummy-coded) predictor set, evaluated across $k = 3, 5, 10, 15, 20, 25, 30$.

k	3	5	10	15	20	25	30
Test RMSE ($)	17.98	17.38	16.73	16.65	16.45	16.38	16.47

Table-6: KNN regression test RMSE across candidate values of k.

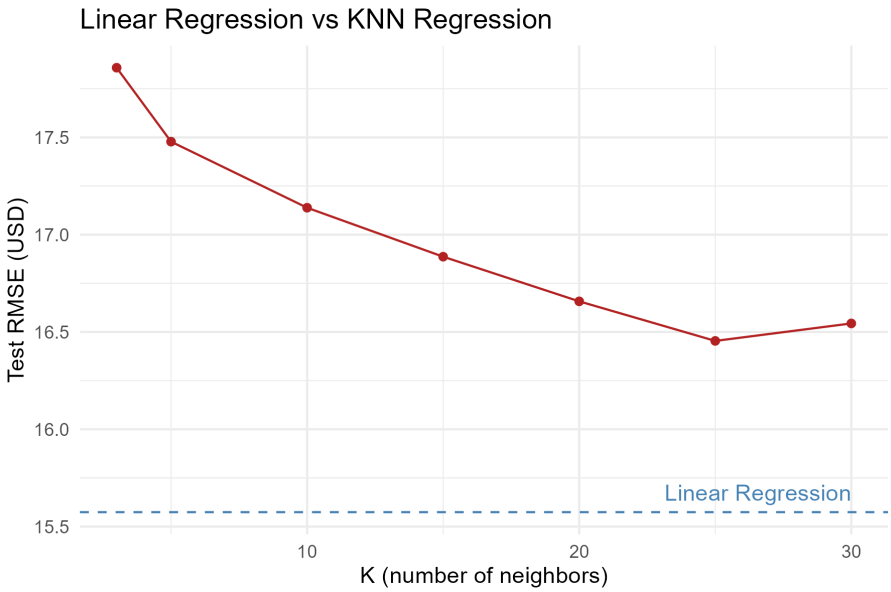

Figure-5: Test RMSE: Linear Regression (dashed reference line) vs KNN Regression across k.

Model	**Test RMSE ($)**
Linear Regression	15.57
KNN Regression ($k = 25$, best)	16.38

Table-7: Predictive accuracy comparison: Linear Regression vs best KNN Regression.

KNN test error falls as k increases from 3 up to 25, then ticks back up slightly at $k = 30$, giving a best result of $16.38 at $k = 25$. The linear regression model outperforms every KNN configuration tested, achieving a test RMSE of $15.57, roughly 5% lower error than the best KNN result. This is consistent with monthly charges being generated by a largely additive, tiered pricing structure that a linear model captures directly, whereas KNN must approximate that structure indirectly through local averaging.

4.3 Summary of Model Comparisons

Results are mixed on this seed-52, 997-customer sample dataset analysis. For classification, LDA edges out logistic regression on this particular test split (80.33% vs. 79.33% accuracy and a meaningfully lower false negative rate), though the difference is modest and likely attributable to sampling variability at this sample size rather than a systematic advantage for either method. For the continuous outcome, linear regression clearly and consistently outperforms KNN regression at every neighborhood size tested, reinforcing that monthly charges are well described by the additive, tiered structure the linear model assumes. Overall, the parametric models remain strong, defensible choices for this dataset; for classification specifically, this particular result is a reminder to consider running the comparison across multiple random splits (e.g., cross-validation) before treating either classifier as definitively superior.